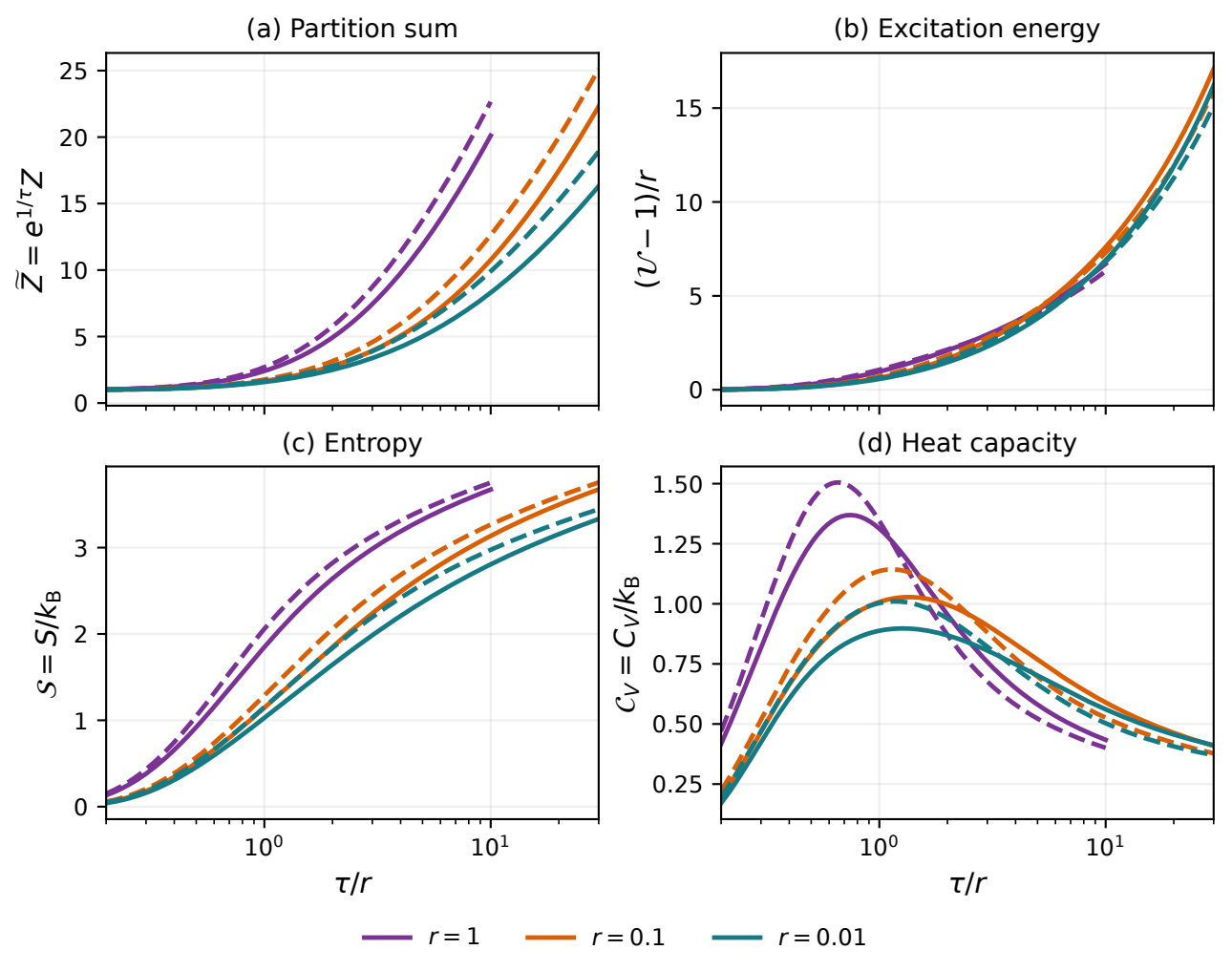
(a) Partition sum
$\widetilde{Z} = e^{1/\tau} Z$
(b) Excitation energy
$(\mathcal{U} - 1)/r$
(c) Entropy
$\mathcal{S} = S/k_B$
τ/r
(d) Heat capacity
$\mathcal{C}_V = C_V/k_B$
τ/r
$r = 1$
$r = 0.1$
$r = 0.01$